

Introduction to Operating Systems

A software system acts as an intermediary between users/applications and computer hardware. It manages hardware resources and provides services to programs and users.

Key functions of an operating system are:

- **Resource Management** - Allocates CPU, memory, and I/O resources
- **Security** - Enforces user authentication, file permissions, and access control
- **Abstraction** - Provides a consistent interface between hardware and software
- **User Interface** - Offers command line interfaces (CLI) or graphical user interface (GUI)